Student Performance Prediction System

Project Report

Submitted by

Aditya Shinde

Project

Student Performance Prediction System

Duration

2 Weeks

Technology

Python & Machine Learning

1. Introduction

The Student Performance Prediction System is a beginner-level Machine Learning project designed to analyze student-related data and predict academic performance.

The project uses factors such as study hours, attendance, previous academic score, sleep hours, extracurricular participation, and internet access to predict a student's **final exam score**.

The project was developed using Python and standard Machine Learning libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

2. Project Objective

The main objective of this project is to build a Machine Learning model that can analyze student-related information and predict a student's final academic performance.

The project aims to:

- Analyze student academic and lifestyle data.
- Clean and prepare the dataset.
- Identify patterns in the data using visualizations.
- Select relevant features for prediction.
- Train and compare Machine Learning models.
- Evaluate the models' predictions.
- Predict the performance of a new student.

3. Technologies Used

- **Python** — Programming language used for the project.
- **Pandas** — Used for data loading, cleaning, and analysis.
- **NumPy** — Used for numerical operations.
- **Matplotlib & Seaborn** — Used to create graphs and visualize data.
- **Scikit-learn** — Used for Machine Learning: data splitting, model training, and evaluation.
- **Jupyter Notebook / Google Colab** — Used to write and execute the Python code.

4. Dataset Description

The dataset contains information about 505 student records, covering academic and lifestyle-related factors. The dataset includes the following columns:

Column	Description
study_hours	Average daily study hours

attendance	Attendance percentage
previous_score	Score obtained in the previous exam/term
sleep_hours	Average daily sleep hours
extracurricular	Whether the student does extracurricular activities (Yes/No)
internet_access	Whether the student has internet access at home (Yes/No)
final_score	Target variable — predicted final exam score

5. Project Workflow

The project follows these steps:

Data Collection → Data Cleaning → Exploratory Data Analysis → Feature Selection → Train-Test Split → Model Training → Model Evaluation → Prediction System

6. Data Cleaning

Data cleaning was performed before training the Machine Learning model. The dataset was checked for missing values and duplicate records. 20 missing values and 5 duplicate rows were found in the raw data (505 rows).

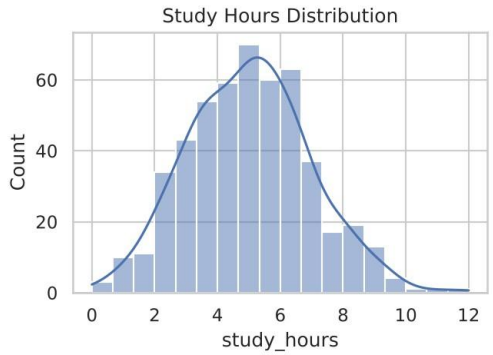
- Checked for missing values — found in the attendance and sleep_hours columns.
- Filled missing numeric values using the column median (robust to outliers).
- Checked for and removed duplicate records.
- Checked the data types of all columns.
- Converted text-based categorical values (extracurricular, internet_access) into numeric valuesusing Label Encoding.

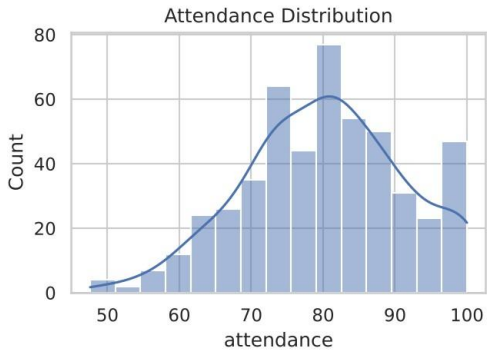
After cleaning, the dataset contained **500** clean, usable rows.

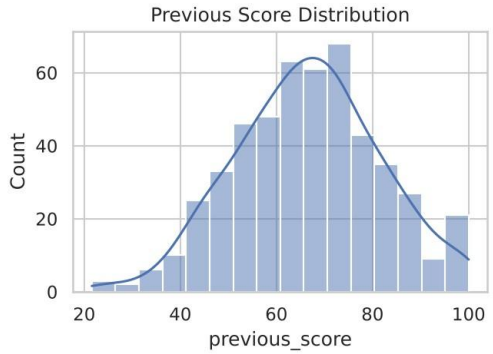
7. Exploratory Data Analysis

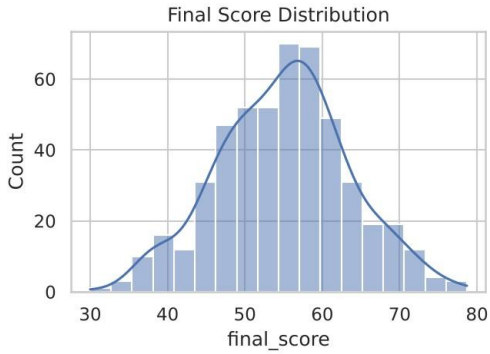
Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns between different student-related factors. The following visualizations were created:

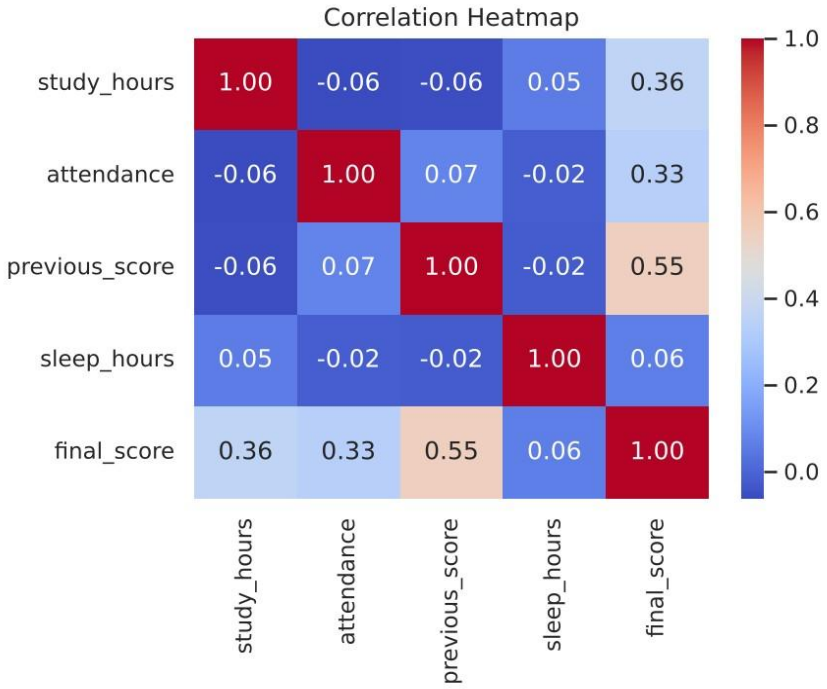
- Distributions of study hours, attendance, previous score, and final score.
- A correlation heatmap between all numeric features.
- Study hours vs. final score and attendance vs. final score.


Study Hours Distribution
Count
study_hours
0
20
40
60
0
2
4
6
8
10
12

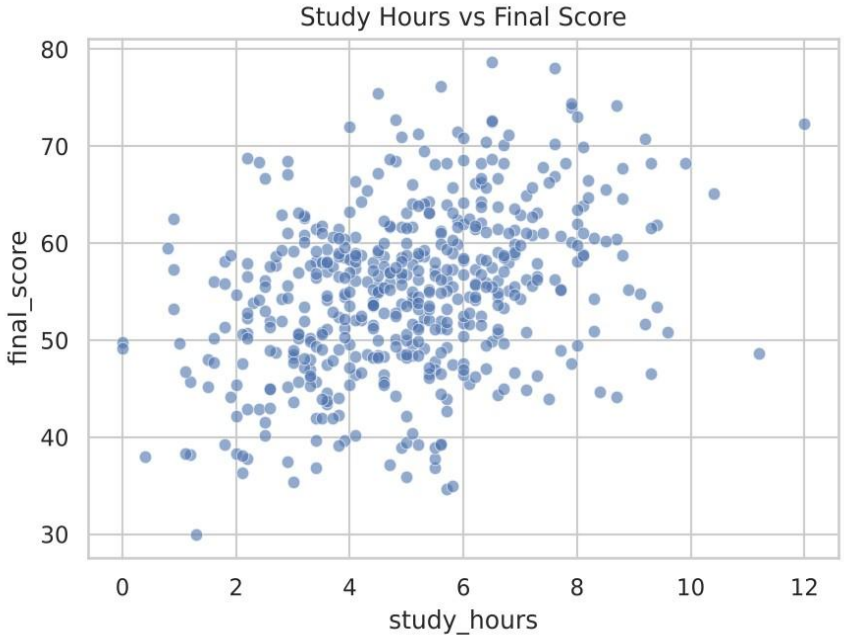

Attendance Distribution
Count
attendance
0
20
40
60
80
50
60
70
80
90
100

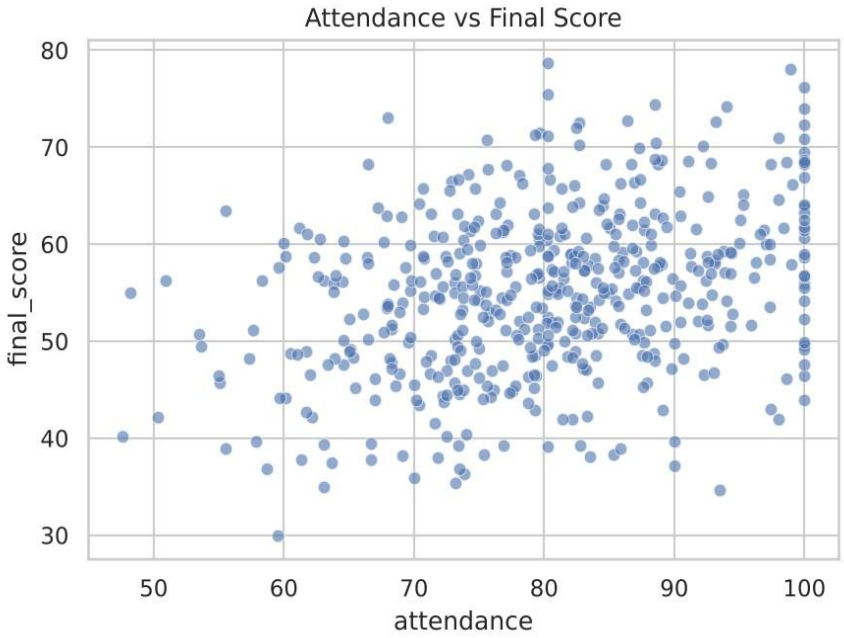

Previous Score Distribution
Count
previous_score
0
20
40
60
20
40
60
80
100


Final Score Distribution
Count
final_score
0
20
40
60
30
40
50
60
70
80


Correlation Heatmap
study_hours
attendance
previous_score
sleep_hours
final_score
1.00
-0.06
-0.06
0.05
0.36
-0.06
1.00
0.07
-0.02
0.33
-0.06
0.07
1.00
-0.02
0.55
0.05
-0.02
-0.02
1.00
0.06
0.36
0.33
0.55
0.06
1.00
study_hours
attendance
previous_score
sleep_hours
final_score
1.0
0.8
0.6
0.4
0.2
0.0

The correlation analysis showed that **previous_score** had the strongest correlation with final score (r = 0.55), followed by **study_hours** (r = 0.36) and **attendance** (r = 0.33). **sleep_hours** showed a weaker correlation (r = 0.06).





8. Feature Selection

The following features were selected to predict student performance:

- Study Hours

- Attendance
- Previous Score
- Sleep Hours
- Extracurricular (encoded)
- Internet Access (encoded)

The target variable is **final_score**.

These features were used as inputs for training the Machine Learning model.

9. Train-Test Split

The dataset was divided into two parts: training data and testing data.

- **80% of the data** was used for training the Machine Learning models.
- **20% of the data** was used for testing the models.

The training data was used to teach the models, while the testing data was used to check how well each model performed on unseen data.

10. Model Training

Three regression models were trained and compared: **Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**.

The models were trained using the training dataset. The selected student features were provided as inputs, and **final_score** was used as the target variable.

```
Model trained successfully!
```

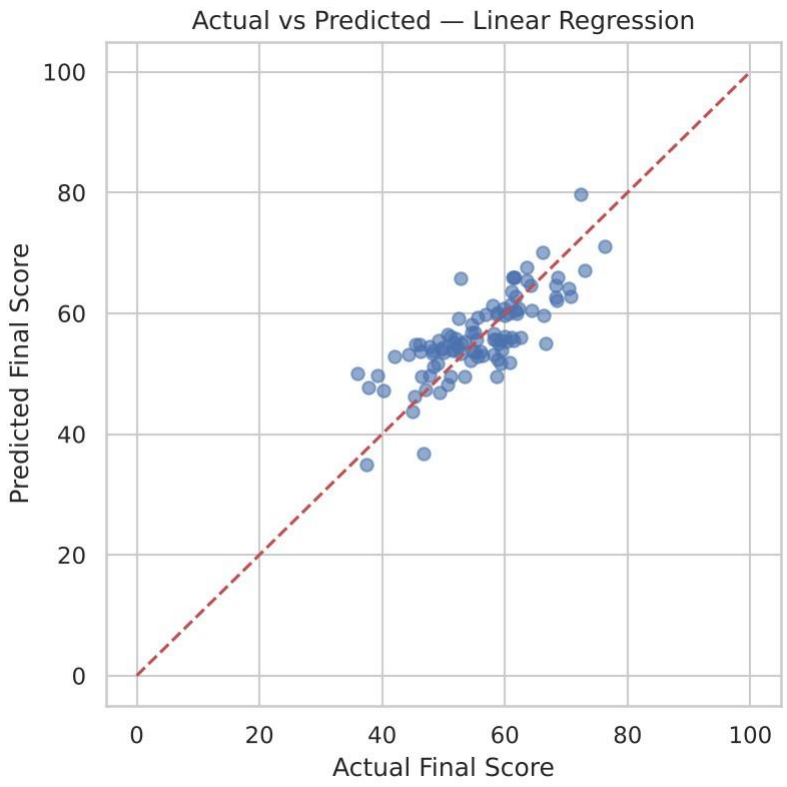
11. Model Evaluation

Each trained model was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R^2 Score**. The predictions made by each model were compared against the actual final scores in the testing dataset.

Model	MAE	RMSE	R2 Score
Linear Regression	4.34	5.27	0.587
Random Forest	4.87	5.99	0.468
Decision Tree	5.58	7.04	0.264

Linear Regression performed best overall, with the lowest error and highest R^2 score (MAE = 4.34, RMSE = 5.27, R^2 = 0.587).

To confirm this result was stable and not specific to one train-test split, 5-fold cross-validation was applied to the best model, giving a mean R^2 score of **0.597** across folds [0.506, 0.712, 0.687, 0.594, 0.488].



12. Prediction Result

The trained model was used to predict the performance of a new student.

The sample student had the following information:

- Study Hours: 6
- Attendance: 85%
- Previous Score: 70
- Sleep Hours: 7
- Extracurricular: Yes
- Internet Access: Yes

The model predicted the student's final score as:

```
Predicted Final Score: 61.86
```

13. Conclusion

The Student Performance Prediction System demonstrates how Machine Learning can be used to analyze student-related data and predict academic performance.

The project included data cleaning, exploratory data analysis, feature selection, model training, and model evaluation across three regression models. **Linear Regression** was selected as the final model based on its evaluation metrics.

The system was also used to generate a prediction for a new student based on their academic and lifestyle-related information.

This project helped in understanding the complete workflow of a Machine Learning project using Python and Scikit-learn.

14. Project Limitations

- The model's performance is based on a limited testing dataset.
- The prediction should not be considered a reliable real-world assessment of studentperformance.

15. Future Improvements

The project can be improved in the future by:

- Using a larger real-world student dataset.
- Testing additional Machine Learning algorithms (e.g. Gradient Boosting, XGBoost).
- Tuning hyperparameters to further improve prediction accuracy.
- Creating a simple web interface (e.g. using Streamlit) for the prediction system.
- Allowing users to enter student details and receive predictions automatically.

16. Project Files

The project repository contains the following files:

- **Student_Performance_Prediction.ipynb** — Jupyter Notebook containing the complete Python code and analysis.
- **student_data.csv** — Dataset used for the project.
- **PROJECT_REPORT.pdf** — Documentation of the project, methodology, results, and conclusions.
- **README.md** — Repository overview and instructions to run the project.